

BOWDET: AN INTERACTIVE TOOLKIT FOR AUDIO-BASED BOW-CHANGE DETECTION

Haotian Yuan

Royal College of Music
London, United Kingdom
yuanhaotianv@gmail.com

Yushi Su

Renmin University of China
Beijing, China
soyeosi108@ruc.edu.cn

ABSTRACT

We present BowDET, a two-stage audio-based toolkit for detecting bow changes in bowed string performance. Because bow changes appear as region-like acoustic transitions rather than sharp onsets, BowDET separates high-recall boundary proposal from bow-change classification. Stage 1 proposes candidates from log-mel spectral differences, while Stage 2 classifies them using a compact CNN fused with boundary-level acoustic features. We additionally introduce a mel-spectrogram-assisted annotation protocol designed to address systematic audio-only labelling latency. On a leave-one-performer-out benchmark spanning violin, viola, and cello (11 performers, 20 recordings, 62.5 min, and 2,516 bow changes), BowDET achieves $\text{IoU}@0.1 \text{ F1} = 66.1 \pm 0.7\%$, outperforming onset-detection and frame-level baselines by more than six percentage points. Through the browser-based demo, users can upload WAV or MP3 recordings, inspect detected boundaries on waveform and spectrogram views, refine predictions as editable annotations, and export timestamped events. Code, pretrained weights, and the interactive demo are publicly available.

1. INTRODUCTION

Bow changes — transitions between down-bow and up-bow strokes — are among the most fundamental gestural events in bowed string performance, useful for practice feedback, bowing annotation, and score–performance alignment [1]. Automatic detection has mostly relied on sensors or video [3, 4], which limit deployment to controlled environments; audio-only methods remain underexplored, and standard onset detectors [2, 6] target sharp note beginnings rather than the gradual, region-like transitions that characterise a bow change.

We present **BowDET**, an open-source system that treats bow changes as boundary regions rather than point events, separating *where* an acoustic boundary occurs from

whether it is a bow change. We also introduce a mel-assisted annotation protocol designed to address systematic audio-only labelling latency. We evaluate BowDET on a cross-instrument, cross-performer audio-only benchmark for bow-change detection, and release a working demo and public implementation.

2. BOWDET SYSTEM

2.1 Dataset and Annotation

The benchmark comprises 20 recordings from 11 performers on violin, viola, and cello (62.5 min, 2,516 bow changes), spanning repertoire from Bach to Penderecki. Evaluation uses strict leave-one-performer-out (LOPO) cross-validation.

Initial audio-only annotations lagged the visible spectral transition by ~ 200 ms, marking perceived completion rather than onset. Our mel-assisted protocol aligns labels to the onset of the visible broadband transition, verified by slowed playback; a second annotator’s independent labels on five recordings give inter-annotator $\text{IoU}@0.1 \text{ F1} = 83.9\%$, providing preliminary evidence for annotation consistency.

2.2 Two-Stage Boundary Modelling

Stage 1 (proposal). From a 64-band log-mel spectrogram (22,050 Hz, hop 512), we compute the frame-wise spectral-difference norm, smooth it, and pick peaks above the 75th percentile as boundary candidates, achieving 0.97 mean recall under $\text{IoU}@0.1$.

Stage 2 (classification). Each candidate is classified using a compact CNN over a 0.74 s log-mel window, fused with a small MLP over 15 hand-crafted boundary features (local spectral-difference statistics, band-wise energy changes, spectral flux). The fused representation feeds a sigmoid output trained with class-balanced binary cross-entropy.

The two-stage design separates boundary localisation from bow-change classification. Removing the hand-crafted features reduces IoU F1 from 66.1% to 54.9%, while replacing Stage 1 with uniform candidate sampling reduces it to 56.6%, confirming the contribution of both components.



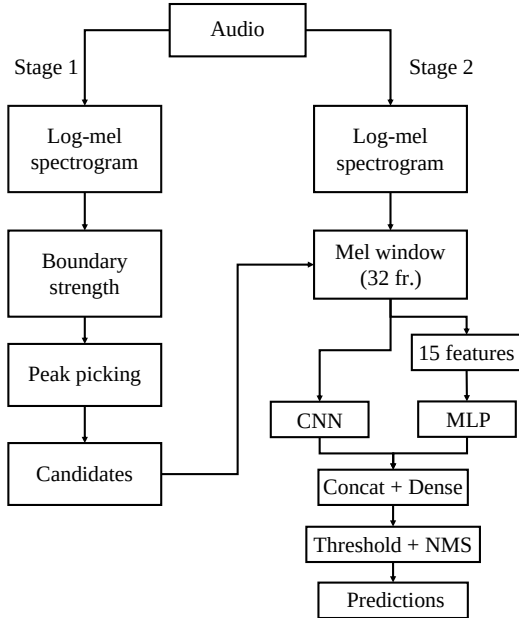


Figure 1. BowDET overview: Stage 1 proposes boundary candidates from log-mel spectral differences; Stage 2 classifies each candidate with a CNN + hand-crafted-feature MLP.

Table 1. LOPO results (IoU@0.1 F1). BowDET is reported as mean over 3 seeds. Inter-annotator agreement on five recordings was 83.9%.

System	IoU@0.1 F1 (%)
Librosa onset	58.7
SuperFlux	59.4
Frame-level CNN	59.6
BowDET (proposed)	66.1 ± 0.7

3. EVALUATION AND DEMO

3.1 Evaluation Protocol

We evaluate under strict LOPO cross-validation using IoU F1: each event is converted into a ± 250 ms region and a match requires $\text{IoU} \geq 0.1$, suited to the region-like, timing-uncertain nature of bow changes. All results are mean $\pm$ SD over three random seeds.

3.2 Results

Table 1 reports LOPO results. The librosa onset detector [5] and SuperFlux [6] achieve high recall but low precision on gradual mid-note transitions; a frame-level CNN without explicit boundary proposals likewise underperforms. BowDET improves on all comparable baselines by more than 6 points; the gap to annotator agreement reflects the inherent ambiguity of bow-change annotation. Figure 2 shows qualitative examples: BowDET tracks clear bow changes closely (a), while dense textures and sparse passages remain the main failure modes (b, c); proposal alone over-predicts compared to the full model (d).

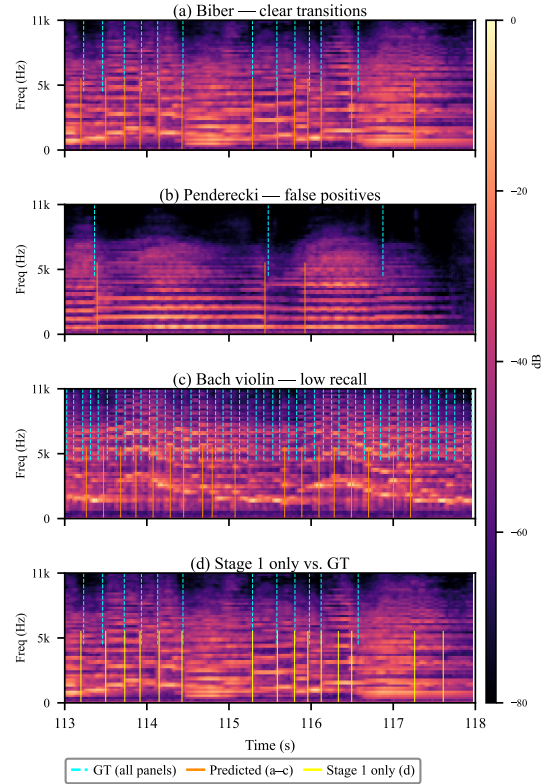


Figure 2. Qualitative examples. Cyan dashed: ground truth; predicted boundaries overlaid. (a) Clear transitions detected accurately. (b) Dense texture causes false positives. (c) Sparse passages yield low recall. (d) Stage 1 alone vs. the full model.

3.3 Demo Availability

BowDET is released as `bowdet`, an open-source, pip-installable toolkit (`pip install bowdet`) with a pretrained model on Hugging Face. In the interactive demo, users upload a recording in WAV or MP3 format; the system displays the log-mel spectrogram overlaid with predicted bow-change lines, which can be accepted or corrected as an editable annotation layer and exported as timestamps. Code and pretrained models: <https://github.com/Yuan-618/bowdet>; demo: <https://huggingface.co/spaces/Haotian-Yuan/bowdet>.

4. CONCLUSION

BowDET demonstrates that treating bow changes as boundary regions rather than point events offers a simple yet effective formulation for audio-only bow-change detection, providing a practical starting point for future work on bowed-string performance analysis.

5. AI USAGE STATEMENT

Generative AI tools (ChatGPT and Claude) were used to assist with language editing, paraphrasing, and translation during the preparation of this paper. All AI-assisted con-

tent was reviewed, revised, and verified by the authors, who take full responsibility for the final manuscript.

6. REFERENCES

- [1] M. Müller, *Fundamentals of Music Processing: Audio, Analysis, Algorithms, Applications*. Springer, 2015.
- [2] J. P. Bello, L. Daudet, S. Abdallah, C. Duxbury, M. Davies, and M. B. Sandler, “A tutorial on onset detection in music signals,” *IEEE Trans. Speech Audio Process.*, vol. 13, no. 5, pp. 1035–1047, 2005.
- [3] D. Dalmazzo and R. Ramirez, “Bowing gestures classification in violin performance: A machine learning approach,” *Frontiers in Psychology*, vol. 10, p. 344, 2019.
- [4] S.-W. Sun, B.-Y. Liu, and P.-C. Chang, “Deep learning-based violin bowing action recognition,” *Sensors*, vol. 20, no. 20, p. 5732, 2020.
- [5] B. McFee *et al.*, “librosa: Audio and music signal analysis in Python,” in *Proc. SciPy*, Austin, TX, 2015, pp. 18–24.
- [6] S. Böck and G. Widmer, “Maximum filter vibrato suppression for onset detection,” in *Proc. DAFx*, Maynooth, Ireland, 2013, pp. 190–196.